

L4.3 Latency and Availability Attacks for AD

Quiz ANS

1. What is the central difference between integrity attacks (L4.1, L4.2) and availability / latency attacks (L4.3)?

- A) Latency attacks often leave the output unchanged but late, so accuracy monitors stay silent
- B) Latency attacks always change the predicted class, so accuracy monitors fire at once
- C) Latency attacks work only on cameras, while integrity attacks work on every sensor
- D) Latency attacks require that the attacker knows the ground-truth labels of the scene

ANS: A) Latency attacks often leave the output unchanged but late, so accuracy monitors stay silent

2. A perception pipeline is delayed by 200 ms while the vehicle drives at 30 m/s. How far does it travel in that extra time?

- A) 0.6 m
- B) 6 m
- C) 60 m
- D) 15 m

ANS: B) 6 m

3. What is 'intermediate-work amplification'?

- A) An input that is much larger in size than usual, such as a huge high-resolution image
- B) A normal-sized input drives many more boxes, tracks or tokens through a costly stage
- C) A flood of requests that exhausts the bandwidth and the request queue of a server
- D) An increase of the learning rate of the victim model during its online training

ANS: B) A normal-sized input drives many more boxes, tracks or tokens through a costly stage

4. The survey defines the amplification factor as $AF = M(x_{adv}) / \text{median } M(x_0)$. What must be held constant between the attacked and the benign measurements?

- A) The attacker's perturbation budget and the number of attack iterations
- B) The random seed that was used for generating the adversarial examples
- C) The model, batch size, hardware and concurrency
- D) The operating system version that was installed on the attacker's machine

ANS: C) The model, batch size, hardware and concurrency

5. Why does the survey prefer the absolute deadline-miss increase ($p_{attack} - p_{clean}$) to a ratio of miss rates?

- A) Ratios cannot be computed correctly in floating-point arithmetic on a modern GPU
- B) Absolute numbers are easier to plot in a figure than ratios of two numbers are
- C) Clean systems have a near-zero miss rate, so a ratio is undefined or meaningless
- D) Ratios cannot be compared across different GPUs and different software versions

ANS: C) Clean systems have a near-zero miss rate, so a ratio is undefined or meaningless

6. A job has absolute deadline $d = 45$ ms and completes at $f = 60$ ms. What are its slack $S = d - f$ and tardiness $\tau = \max(0, f - d)$?

- A) $S = 15$ ms, $\tau = 0$ ms
- B) $S = 105$ ms, $\tau = 60$ ms
- C) $S = -15$ ms, $\tau = 0$ ms
- D) $S = -15$ ms, $\tau = 15$ ms

ANS: D) $S = -15$ ms, $\tau = 15$ ms

7. An EOS-delaying prompt, a no-EOS backdoor and an EOS-suppressing bit flip all lengthen the generated text. In the survey's terms, how do they relate?

- A) Three different bottlenecks that are reached through one common delivery channel
- B) Same bottleneck (output length), three delivery channels; one hard token cap bounds all

C) Three attacks that cannot be bounded by one common control mechanism

D) Not latency attacks at all, because every one of them changes the final answer

ANS: B) Same bottleneck (output length), three delivery channels; one hard token cap bounds all

8. Why do many small, spread-out phantom boxes make greedy NMS slow?

A) They overlap nothing, so almost none is suppressed and the work nears n^2 comparisons

B) They raise the image resolution, so that the detector head needs much more compute

C) They force the detector to run a second time on the same input image again

D) They are suppressed at once, but sorting all of them is the expensive part

ANS: A) They overlap nothing, so almost none is suppressed and the work nears n^2 comparisons

9. In the Groundswell example, a $188\times$ rise in NMS candidates gives only a $2.7\times$ rise in end-to-end latency. Why?

A) NMS is a small part of the benign latency, so the other stages dilute the effect

B) The attack was run on a slower GPU than the one that was used for the clean runs

C) End-to-end latency is always measured in frames and not in milliseconds

D) The tracker removed most of the candidates before NMS could process them at all

ANS: A) NMS is a small part of the benign latency, so the other stages dilute the effect

10. What does the survey conclude about the evidence level of physical latency attacks (e.g., SlowPerception, DetStorm)?

A) All of them were demonstrated end to end on public roads with real traffic

B) None of them was ever evaluated in a simulator or on target hardware

C) Physical delivery is shown only in pieces; no entry is a full on-road deployment

D) Only digital attacks were ever measured on real embedded hardware

ANS: C) Physical delivery is shown only in pieces; no entry is a full on-road deployment

11. What is a key finding of the EVADE study (Monteuuis et al., NeurIPS 2025) on latency NMS attacks?

A) Every NMS attack transfers unchanged to all detectors and all hardware platforms

B) GPU NMS can always be pushed beyond a 500 ms budget by a small patch

C) None of the three tested defenses reduced the effect on post-NMS counts

D) Attacks do not transfer across YOLO versions or sizes, and universal patches fail across datasets

ANS: D) Attacks do not transfer across YOLO versions or sizes, and universal patches fail across datasets

12. What is the key difference between the NMS bottleneck and the tracker bottleneck?

A) The tracker runs before the detector, so it cannot see any of the phantom boxes

B) NMS keeps its state across frames, while the tracker is reset in every frame

C) The cost of the tracker is always linear in the size of the input image

D) The tracker has memory: unmatched false tracks survive for frames, so work accumulates

ANS: D) The tracker has memory: unmatched false tracks survive for frames, so work accumulates

13. Why does a per-frame cap on detections not stop the tracker-pool accumulation of SlowTrack?

A) It bounds one dimension of the $M \times N$ cost, not the persistent pool of tracks

B) Caps of this kind only work in simulation and never on real embedded hardware

C) The cap is applied only after the tracker has already finished its full update

D) SlowTrack never creates more than one single false track in each frame

ANS: A) It bounds one dimension of the $M \times N$ cost, not the persistent pool of tracks

14. Exercise: $T = 30$ ms, deadline $D = 45$ ms, the tracker has share $f = 5\%$ and is amplified by $A = 50$ ($A_{\text{eff}} = 1 + f(A - 1)$). What is the new pipeline latency, and is the deadline missed?

A) About 45 ms; no, the deadline is met exactly

B) About 1,500 ms; yes, the deadline is missed

C) About 36 ms; no, the deadline is met

D) About 103.5 ms; yes, the deadline is missed

ANS: D) About 103.5 ms; yes, the deadline is missed

15. Continuing the exercise ($T = 30$ ms, $D = 45$ ms, $f = 5\%$), an admission cap limits the amplification to $A = 5$. What is the result?

- A) $A_{e2e} = 5$, latency 150 ms; the deadline is missed
- B) $A_{e2e} = 3.45$, latency 103.5 ms; the deadline is missed
- C) $A_{e2e} = 1.2$, latency 36 ms; the deadline is met
- D) $A_{e2e} = 0.2$, latency 6 ms; the pipeline would be idle

ANS: C) $A_{e2e} = 1.2$, latency 36 ms; the deadline is met

16. Why does TrackFlood work against detectors without NMS (YOLOv10, RT-DETR)?

- A) These detectors have no detection head, so the tracker receives no input
- B) The detector latency grows by $100\times$, while the tracker stays unchanged
- C) Detectors without NMS cannot be given an adversarial perturbation
- D) The bottleneck moves to the tracker, so NMS-centric defenses have nothing to act on

ANS: D) The bottleneck moves to the tracker, so NMS-centric defenses have nothing to act on

17. What is the difference between an admission cap and an emission cap?

- A) Admission limits what leaves a stage, while emission limits what enters that stage
- B) Both bound the cost of the stage in exactly the same way for every detector
- C) Admission limits work entering a stage; emission (max_det) limits only its output
- D) Emission caps are used only for tokens and admission caps only for boxes

ANS: C) Admission limits work entering a stage; emission (max_det) limits only its output

18. Which control on reasoning length gives a hard guarantee according to the survey?

- A) Runtime termination or token accounting that enforces a configured upper bound
- B) A prompt instruction that politely asks the model to think only briefly
- C) A learned length direction that shortens the answers on average over many prompts
- D) A detector that flags verbose prompts before they reach the reasoning model

ANS: A) Runtime termination or token accounting that enforces a configured upper bound

19. Why does the lecture say that AI-based perception has no WCET (worst-case execution time)?

- A) Neural networks run in constant time on every possible input that they receive
- B) Run time depends on the input, and attacks find high-cost inputs, not proven maxima
- C) WCET analysis is forbidden for machine-learning components by the ISO 26262 standard
- D) GPUs cannot be timed accurately by any measurement tool that is available today

ANS: B) Run time depends on the input, and attacks find high-cost inputs, not proven maxima

20. Two vehicles suffer the same perception delay. Vehicle 1 keeps driving on stale data; vehicle 2 performs a safe stop. According to condition C6 (consequence mapping), what is the likely difference?

- A) Both vehicles crash, because the delay is identical in the two cases
- B) Vehicle 1 may crash, while vehicle 2 loses availability but avoids a collision
- C) Vehicle 2 crashes, because stopping on the road is always an unsafe action
- D) Neither vehicle is affected, because latency attacks do not change any behaviour

ANS: B) Vehicle 1 may crash, while vehicle 2 loses availability but avoids a collision